8 Eksponensial o’sish va kamayish

1-to’plam. *Jadval chiziqlimi yoki eksponensialmi? Qoidasini (y = ...) yozing.*

1. $y = 5,\ 10,\ 20,\ 40$

2. $y = 3,\ 7,\ 11,\ 15$

3. $y = 100,\ 90,\ 81,\ 72.9$

4. $y = 50,\ 45,\ 40,\ 35$

5. $y = 2,\ 6,\ 18,\ 54$

6. $y = 10,\ 12.5,\ 15,\ 17.5$

2-to’plam. *O’sish: tenglama tuzing, so’ng hisoblang.*

1. A town’s population is 200 and grows 5% per year. Find the population after 4 years.

2. An investment of \$1000 grows 6% per year. Find its value after 3 years.

3. A bacteria culture starts at 50 and doubles every hour. Find the count after 5 hours.

4. A city’s population is 8000 and grows 3% per year. Find the population after 2 years.

5. An item costs \$20 and its price increases 10% per year. Find the price after 3 years.

6. A quantity starts at 12 and grows by a factor of 1.5 each step. Find its value after 4 steps.

3-to'plam. *Kamayish: tenglama tuzing, so'ng hisoblang.*

1 A car worth \$15000 depreciates 12% per year. Find its value after 3 years.

2 A 200 g sample decreases 8% per year. Find the amount after 2 years.

3 A population of 5000 declines 4% per year. Find the population after 5 years.

4 100 mg of medication decreases 25% every hour. Find the amount after 3 hours.

5 An 800 L tank loses 10% of its water each day. Find the amount after 2 days.

6 A \$2000 value decreases by a factor of 0.8 each year. Find the value after 3 years.

4-to'plam. *Avval chiziqli/eksponensialligini aniqlang, so'ng hisoblang.*

1 A plant grows 3 cm every week, starting at 10 cm. Height after 6 weeks?

2 A savings account earns 4% interest per year, starting at \$500. Value after 3 years?

3 A phone loses 15% of its charge every hour, starting at 100%. Charge after 2 hours?

4 A pool drains at a constant rate of 50 liters per minute, starting with 2000 liters. Amount after 10 minutes?

5 Bacteria triple every day, starting with 4. Count after 4 days?

6 A runner's distance increases by 2 miles every day, starting at 1 mile. Distance after 5 days?

5-to’plam. *Yarimparchalanish davri (half-life).*

1 160 g of a substance has a half-life of 4 years. How much remains after 12 years?

2 500 mg of a drug has a half-life of 6 hours. How much remains after 18 hours?

3 80 g of an isotope has a half-life of 10 days. How much remains after 30 days?

4 A population halves every 5 years, starting at 6400. Find the population after 20 years.

5 1000 bacteria die off such that the count halves every 3 hours. Find the count after 9 hours.

6-to’plam. *Xatoni toping – buzilgan qadamni ayting, to’g’ri javobni bering.*

1 A city grows 7% per year, starting at 10000. Find the population after 2 years –

“$10000 \times (1.7)^2 = 28900$.”

Xato va to’g’ri javob: .....

2 A sample decreases 20% per year, starting at 500. Find the amount after 2 years –

“$500 \times (1 + 0.2)^2 = 720$.”

Xato va to’g’ri javob: .....

3 A table shows $y = 2, 4, 8, 16$ for $x = 0, 1, 2, 3$ –

“This is linear with common difference 2, so $y = 2 + 2x$.”

Xato va to’g’ri javob: .....

4 A car depreciates by \$1500/year, starting at \$20000 –

“This is exponential: $y = 20000 \times (1 - 0.075)^t$.”

Xato va to’g’ri javob: .....

5 Amount remaining after 3 half-lives, starting at 240 g –

“$240 - 3 \times \frac{1}{2} \times 240 = 240 - 360 = -120$.”

Xato va to’g’ri javob: .....

7-to’plam. *Bonus / qiyin – teskari masala: berilgan natijadan parametrni toping.*

1 A population grows from 200 to 242 in 2 years at a constant annual percent rate. Find the rate.

2 A substance decays to $\frac{1}{4}$ of its original amount after 10 days. Find its half-life.

3 An investment grows from \$1000 to \$1331 in 3 years at a constant annual rate. Find the rate.

4 A car worth \$25000 depreciates to \$20000 after 1 year at a constant percent rate. Find the rate, then predict the value after a total of 2 years.

5 A quantity grows from 3 to 48 over 4 equal steps. Is this linear or exponential? Find the rule.

Javoblar

To'plam	#	Javob
1-to'plam	1	Exponential, $y = 5 \times 2^x$
	2	Linear, $y = 3 + 4x$
	3	Exponential, $y = 100 \times 0.9^x$
	4	Linear, $y = 50 - 5x$
	5	Exponential, $y = 2 \times 3^x$
	6	Linear, $y = 10 + 2.5x$
2-to'plam	1	≈ 243
	2	$\approx$ \$1191.02
	3	1600
	4	≈ 8487
	5	$\approx$ \$26.62
	6	60.75
3-to'plam	1	$\approx$ \$10222.08
	2	$\approx$ 169.28 g
	3	≈ 4077
	4	$\approx$ 42.19 mg
	5	648 L
	6	\$1024
4-to'plam	1	Linear, 28 cm
	2	Exponential, $\approx$ \$562.43
	3	Exponential, $\approx$ 72.25%
	4	Linear, 1500 L
	5	Exponential, 324
	6	Linear, 11 miles

To’plam	#	Javob
5-to’plam	1	20 g
	2	62.5 mg
	3	10 g
	4	400
	5	125
6-to’plam	1	Error – 7% treated as 1.7 instead of 1.07. To’g’ri: $10000 \times 1.07^2 \approx 11449$.
	2	Error – used $(1 + r)$ for a decaying quantity instead of $(1 - r)$. To’g’ri: $500 \times 0.8^2 = 320$.
	3	Error – differences 2, 4, 8 are not constant; ratios 2, 2, 2 are constant, so it’s exponential: $y = 2 \times 2^x$.
	4	Error – a constant dollar amount per year is linear, not exponential. To’g’ri: $y = 20000 - 1500t$.
	5	Error – treated repeated halving as subtraction instead of multiplication. To’g’ri: $240 \times (\frac{1}{2})^3 = 30$ g.
7-to’plam	1	$r = 10\%$
	2	$h = 5$ days
	3	$r = 10\%$
	4	$r = 20\%$; after 2 years, \$16000
	5	Exponential, $y = 3 \times 2^x$

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